

IPSC CORTICAL DIFFERENTIATION

All methods described below should be performed using sterile technique.

Preparation of feeder-free iPSCs culture

1. Coat appropriate sized tissue culture dish with Matrigel. Swirl plate to distribute evenly across surface of dish. Allow Matrigel to set for 1hr at room temperature or in a 37 °C humidified chamber. Note: this will provide enough cells for a full 96 well plate of neural aggregates.

NOTE: We typically thaw 1 vial of iPSC into 1 well of a 6 well (1mL Matrigel per well).

Volumes below are based on 6 well plates.

2. Prepare mTeSR1 feeder-free medium by aseptically adding 100mL of thawed 5x supplement to 400mL Basal Medium. Add 5mL penicillin/streptomycin if needed. Mix thoroughly and warm to room temperature.
3. Thaw iPSCs ($1-2 \times 10^6$ cells/mL) in a 37 °C water bath. Add 1 mL of thawed iPSC to 9 mL of DMEM/F12 in a 15mL conical tube (Do not mix). Centrifuge iPSC at 750 rpm for 3 minutes.
4. Carefully aspirate medium from iPSC pellet. Add 2mL of mTeSR1 supplemented with Rock inhibitor (10 μ M final) - avoid breaking up clumps of suspended cells.
5. Aspirate Matrigel from tissue culture dish. Add 2mL of suspended iPSCs. Place plate in 37°C, 5% CO₂ and 95% humidified chamber, making a "T" motion to ensure even distribution of cells. Change medium with 2mL of fresh mTesR1 medium daily.

NOTE: As iPSC become more confluent, increase mTesR to 3mL per well.

Neural aggregate formation

1. Harvest iPSCs for neural aggregate formation when iPSCs have reached 75-85% confluency. Aspirate medium and rinse with 2mLs of DPBS. Add 1mL of Accutase. Incubate at 37°C for 5 minutes. Gently tap plate to dislodge cells.

Dilute Accutase with 4mL of DMEM/F12 medium and collect cell suspension in 15ml conical tube.

2. Centrifuge cells at 750 rpm for 3 minutes. Then carefully aspirate medium from iPSC pellet. Add 3 mLs of neural induction medium to iPSC pellet. Using a hemacytometer, count iPSCs. Adjust volume of iPSC suspension to 450-650,000 cells/mL using neural induction medium supplemented Rock inhibitor (10 μ M final)
3. Add 100 μ l of iPSC suspension per well to a v-bottom 96-well plate
4. Centrifuge plate at 750 rpm for 3 minutes to sediment iPSC into spheres
5. Incubate cells at 37°C, 5% CO₂ and 95% humidified chamber for 24 hrs. After 24 hrs, carefully remove all medium from well and replace with 100 μ l per well of Neural Induction Medium.

NOTE: Do not disturb or break apart spheres. The spheres are very delicate at this stage.

6. Incubate neurospheres in 96 well plate for 4 additional days. Perform half volume medium changes daily (removed 50ul and replace with 50ul).

Preparation of tissue culture plates for neural rosettes and neural progenitors

1. Coat 3 wells of a 6-well plate with 1ml Poly-L-Ornithine (PLO) for 2hrs or overnight in humidified incubator. Aspirate off PLO and rinse 3 times with DMEM/F12.
2. Dilute laminin to 10 μ g/mL final in cold DMEM/F12 and add 1 mL/well of solution to PLO coated plates. Incubate for 2 hrs or overnight in humidified incubator.

Neural Rosette Formation and Selection

1. On Day 5 of neural aggregate formation, remove media (by pipetting) and carefully wash spheres with 100 μ l of pre-warmed DMEM/F12. Repeat 2 times.

NOTE: Do not break apart spheres. Neural spheres are very delicate at this

- stage. An alternative approach is to remove 50ul of spent media and wash with 50ul DMEM/F12. Add 50ul fresh neural induction media. Transfer 100ul of spheres and media to the new PLO/laminin-coated well. This approach will transfer more dead cells into the new well.
2. Remove the last wash and add 50 µl of neural induction media to each well.
 3. Aspirate laminin from one well of the pre-coated plate. Using 200 µl sterile tips, carefully pipet up spheres from wells using 100 µl volume and transfer thirty-two spheres per well. Repeat above steps for the remaining wells. Incubate cells in 37°C, 5% CO₂ and 95% humidified chamber and distribute evenly by making a “T” motion.
 4. After 24 hrs, examine attached aggregates. Remove medium and replace with 2mls/well fresh neural induction medium daily. NOTE: If some aggregates have not attached, carefully pipet out all medium and replace with 1ml/well fresh neural induction medium. Once 90-100% of aggregates attach, exchange medium daily with 2mls/well neural induction medium.
 5. Monitor spheres daily under microscope for formation of neural rosette structures. Neural rosettes are ready to harvest when spheres have completely flattened and clusters are clearly visible (3-7 days after plating, line dependent).
 6. Harvest neural rosettes by aspirating spent medium. Add 1ml of pre-warmed DMEM/F12 to each well to remove unattached cells (repeat if necessary). Add 1ml of Neural Rosette Selection reagent to each well and incubate for up to 1 hour at 37 °C (check cells at 20 minutes. Cells are typically collected after 30-45 min incubation. Look for rosette structure to be rounding up without the disturbance of other surrounding cells). Carefully remove Neural Rosette Selection reagent with a pipet, being careful not to disturb rosette clusters. Add 1mL DMEM/F12 to each well, then using a p1000 detach rosette clusters by

rinsing over them. Transfer rosette material from 1 well into a 15 mL conical tube for cryopreservation of neural rosettes and from 2 wells into a separate 15 mL conical tube for neural progenitor expansion. Do not triturate clusters.

NOTE: To maintain a pure culture, it is best to leave some rosettes behind rather than collect all of the rosettes and additional cells.

7. Centrifuge rosette clusters at 750 rpm for 3 minutes.

Neural rosette banking

1. Make 2x stock of serum-free neural freezing medium by adding 10 mLs of sterile DMSO to 40 mLs of sterile KOSR into a 50 ml conical tube. Mix by inverting 3-4 times. Store at 4°C for up to 4 weeks.
2. Aspirate supernatant from 15mL conical tube containing 1 well of neural rosette clusters (see Step 6 above). Add 1.5 mLs of neural induction media supplemented with 1.5 mLs of 2x neural freezing medium to the single well collection of neural rosette clusters. Gently mix solution and distribute 1mL into sterile cryovials. Store cryovials at -80 °C for at least 48 hrs then transfer to liquid nitrogen for long-term storage.

Neural progenitor expansion

1. Using the pre-coated PLO-laminin plate (see above), aspirate laminin from 2-3 wells. Remove supernatant from 15mL conical tube containing 2 wells of neural rosette clusters (see Step 7 above). Add appropriate amount of neural induction media (NIM) to achieve a final volume of 2mL per well. Pipet clusters up and down breaking them into 1/4 or 1/5 the size of the rosette cluster.
2. Add 2 mL of neural rosette cluster suspension to each well and incubate at 37°C, 5% CO₂ and 95% humidified chamber for 24 hours.

3. Examine adherent cells under microscope, aspirate off medium and replace with 2 mL of fresh neural induction medium daily for 3-5 days or until they reach ~80% confluent.
4. Make 12 mL of DMEM/F12 supplemented with 10% FBS by adding 2 mL of FBS to 10 mL of DMEM/F12 to inhibit trypsin activity (termed "Complete Media").
5. Remove medium and rinse cells with 1 mL/well of DMEM/F12. Harvest cells by adding 1 mL/well of 0.05% trypsin and incubate at 37°C for 3 minutes. Inhibit trypsin activity by adding 4mL of DMEM/F12 supplemented with 10% FBS to each well. Collect cells in 15mL conical tube. Centrifuge at 750 rpm for 3 minutes.

NOTE: To minimize bubbles and increase cell yield, add 1mL Complete Media to 15mL conical tube. After trypsinization, add 3mL DMEM/F12 to collect cells and transfer to 15mL conical tube.

6. Aspirate supernatant and resuspend in 12 mLs of NIM, pipet at least 3 times to break up large clumps. NOTE: Small clumps and single cells are acceptable to passage.
7. Passage cells to pre-coated PLO-lamin plates (see above) by adding 2 mLs/well of suspended cells for a total of 6 wells.
8. Feed with 2 mLs of fresh NIM daily until they reach 85-95% confluency.
9. IMPORTANT: NPCs are best maintained by passaging at a 1:3 dilution.
Passaging cells that are more dilute will result in slow growth. Cells will typically become confluent after 2-4 days when plated as a 1:3 dilution. If NPCs are observed to slow in their growth, passage cells and replate without dilution.

Neural progenitor banking

1. Upon reaching at least 85% confluency, harvest neural progenitor cells as described above. Perform a cell count (as described above) in 3 mLs of NIM.
2. Add equal volumes of NIM and 2x neural freezing medium to the NPC cell suspension for a final 1×10^6 cells/mL. Gently mix solution and distribute 1mL into sterile cryovials. Store cryovials in Styrofoam containers at -80 °C for 48 hrs and then transfer to liquid nitrogen for long-term storage.

Thawing Rosettes and NPC

1. Aliquot 9 mLs of pre-warmed DMEM/F12 into a 15 ml conical tube
2. Remove cells from liquid nitrogen and thaw in 37°C water bath for approximately 30 seconds or until a small ice clump remains
3. Add freshly thawed cells into the 9 mLs of pre-warmed DMEM/F12 and mix gently by gently shaking tube 3 times. Avoid breaking up clumps of cells.
4. Centrifuge cells at 750 rpm for 3 minutes. Aspirate supernatant.
5. Resuspend in 2 mLs of NIM and plate into one well of a 6-well plate. Note: If cell viability count is necessary, after resuspension remove 10 µl into a separate microfuge tube for cell viability count

Differentiation of NPC into cortical neurons

1. IMPORTANT: To generate cortical neurons, NPCs must be at passage 4 or lower at the time of plating for terminal differentiation. NPCs that are beyond passage 4 may exhibit higher densities of astrocyte contamination or inefficient neuronal differentiation.
2. Cells were plated at an appropriate density (e.g.: 150K/well in 12-well plate; 75- 50K/well for 48 well plates; 30K/well for 8-well chamber slides) in NIM on pre-coated PLO/laminin plates (see 2.3).

3. After 24hrs, medium was replaced with cortical neuron differentiation medium
(Neurobasal medium, 1x B27, 20ng/mL BDNF, 20ng/mL GDNF, 0.5mM cAMP,
1% Glutamax, 1% penicillin/streptomycin)
4. Cells were fed every 2-3 days for 30 days

Materials

Matrigel (BD Biosciences BD354277)

mTesR1 (Stem Cell Technologies 5870)

STEMdiff Neural Induction Media (NIM; Stem Cell Technologies 5831)

Accutase (Fisher AT104)

DMEM/F12 (Invitrogen 11039021)

FBS (Invitrogen 26140079)

Knockout Serum Replacement (KOSR; Invitrogen 10828-028)

DMSO (Sigma D2650)

PLO (Sigma P4957)

Rock Inhibitor (Y-27632; Fisher 50811206)

Laminin (Sigma L2020)

STEMdiff Neural Rosette Selection Reagent (Stem Cell Technologies

5832) BDNF (Sigma SR3014)

GDNF (Sigma G1777)

Dibutyryl cAMP (D0260)